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 AI Insights

COVID-19 AI Analytics Dashboard

HealthGuard Analytics Pvt. Ltd.

 Project Overview

This project analyzes COVID-19 patient records to provide valuable healthcare insights using Machine Learning and Artificial Intelligence.

The dashboard helps healthcare authorities understand:

- Patient demographics
- Infection trends
- Recovery analysis
- Country-wise statistics
- Machine Learning predictions
- AI generated healthcare insights



Dashboard

Explore

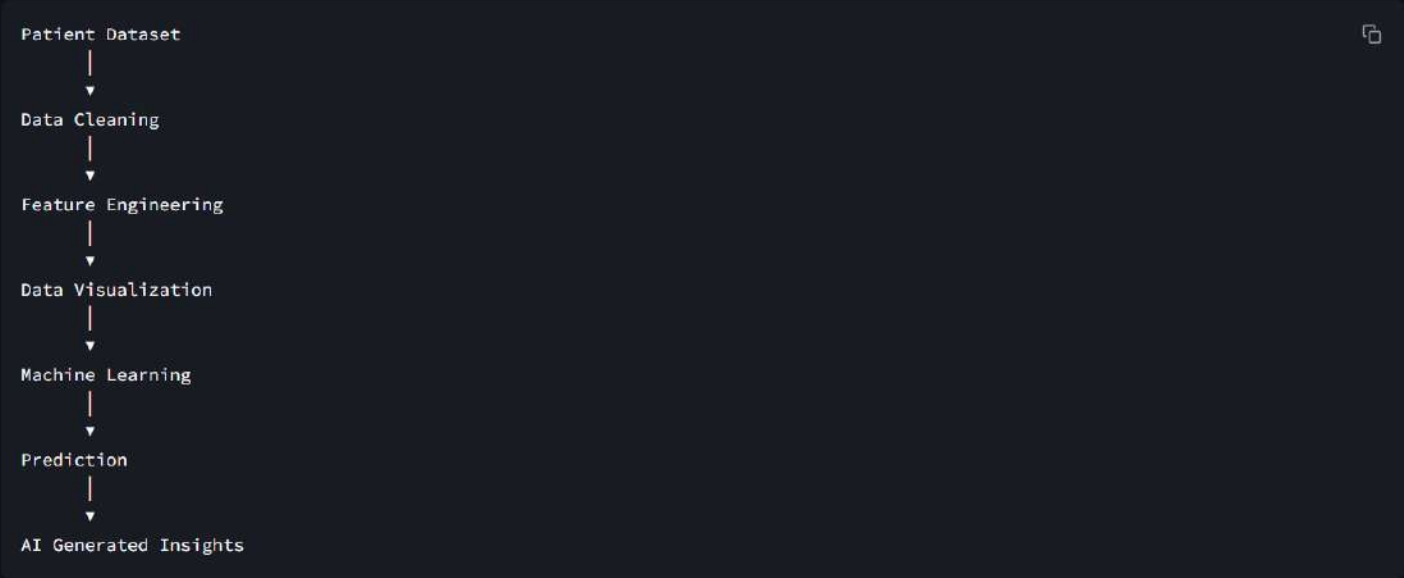
- ✓ Dataset
- ✓ Interactive Graphs
- ✓ Infection Trends
- ✓ Country Analysis
- ✓ Recovery Statistics



AI Features

- ✓ ML Prediction
- ✓ Random Forest Model
- ✓ AI Report Generator
- ✓ Feature Importance
- ✓ Download Report

⚙️ Project Workflow



 Developed By

Akshey Sehgal

B.Tech Robotics & AI

Machine Learning | Data Science | AI

Project for Company Placement



AI Powered COVID-19 Insights

Dataset Preview

id	sex	birth_year	country	region	group	infection_reason	infection_order	infected_by	contact_number	confirmed_date	released_date	deceased
1	female	1984	China	filtered at airport	None	visit to Wuhan	1	None	45	2020-01-20	2020-02-06	None
2	male	1964	Korea	filtered at airport	None	visit to Wuhan	1	None	75	2020-01-24	2020-02-05	None
3	male	1966	Korea	capital area	None	visit to Wuhan	1	None	16	2020-01-26	2020-02-12	None
4	male	1964	Korea	capital area	None	visit to Wuhan	1	None	95	2020-01-27	2020-02-09	None



Executive Strategic Planner



Draft Automated Public Health Strategy Report

Strategic Document Compiled Successfully!

Epidemiological Analysis Report: Early-Stage Infectious Disease Outbreak ↔

Prepared by: Senior Healthcare Data Scientist

Target Audience: Public Health Officials, Clinical Operations Leadership, and Epidemiological Task Force

Data Scope: 8,424 patient records (14 features)

1. Executive Summary

This report provides a rigorous data science evaluation of an active, rapidly expanding infectious disease outbreak. Based on an analysis of 8,424 patient records, the outbreak is heavily concentrated in South Korea (99.8% of cases), with minor initial introductions from China and Mongolia.

The primary clinical finding is that **99.0% of all confirmed patients (8,341 individuals) are currently in isolation**. Only 56 patients have been released, and 26 deaths have been recorded. This extreme skew toward isolation indicates that the outbreak is in a highly active, exponential acceleration phase. The low number of resolved cases (recoveries or deaths) is a classic epidemiological lagging indicator; it reflects the novelty of the surge rather than a low-virulence profile.

Alarming, the dataset is characterized by critical metadata omissions, with missingness rates exceeding 92% for key demographic, geographic, and contact-tracing features. This "data fog" indicates that local public health infrastructure and contact-tracing teams have been overwhelmed by the sheer velocity of the transmission. Immediate operational pivoting is required to standardize data capture, prioritize clinical resource allocation, and transition from manual contact tracing to high-throughput digital surveillance.

2. Key Transmission Findings

An analysis of transmission dynamics reveals a critical vulnerability: the systematic breakdown of contact-tracing and epidemiological link tracking.

The Epidemiological "Data Fog" (Missingness Analysis)

The missing value log serves as a proxy metric for public health capacity. When case volumes outpace human resource capacity, detailed epidemiological investigations are the first systems to fail.

Feature	Missing Values	Percentage Missing
id	0	0.00%
confirmed_date	0	0.00%
country	0	0.00%
state	0	0.00%
sex	7,787	92.44%
region	7,813	92.75%
birth_year	7,839	93.06%
infection_reason	8,163	96.90%
group	8,271	98.18%
infected_by	8,299	98.52%
contact_number	8,359	99.23%
infection_order	8,353	99.16%

Insights from the Transmission Metrics

- **Complete Loss of Transmission Chains:** With `infected_by` missing in 98.52% of cases and `infection_order` missing in 99.16% of cases, we have virtually no visibility into the transmission hierarchy. We cannot reliably calculate the effective reproduction number (R_t) using generation-interval methods.
- **Contact Tracing Paralyzed:** The `contact_number` (tracking the number of close contacts a patient had) is missing for 99.23% of records. This indicates that contact tracing was either abandoned due to volume or could not be logged fast enough, meaning thousands of potentially exposed individuals remain untracked in the community.
- **Unidentified Cluster Vectors:** The `group` column (98.18% missing) and `infection_reason` (96.90% missing) indicate that super-spreading events (e.g., call centers, religious gatherings, or hospitals) are not being systematically categorized, preventing targeted cluster-containment strategies.

3. Regional Risk Assessment

The geographic distribution of the outbreak is highly skewed, demanding localized containment strategies with international vigilance.

Country-Level Distribution (N = 8,424)

Korea:	8,405 cases	(99.77%)
China:	16 cases	(0.19%)
Mongolia:	2 cases	(0.02%)
Noise/Artifact:	1 case	(0.01%)

Domestic Risk (South Korea)

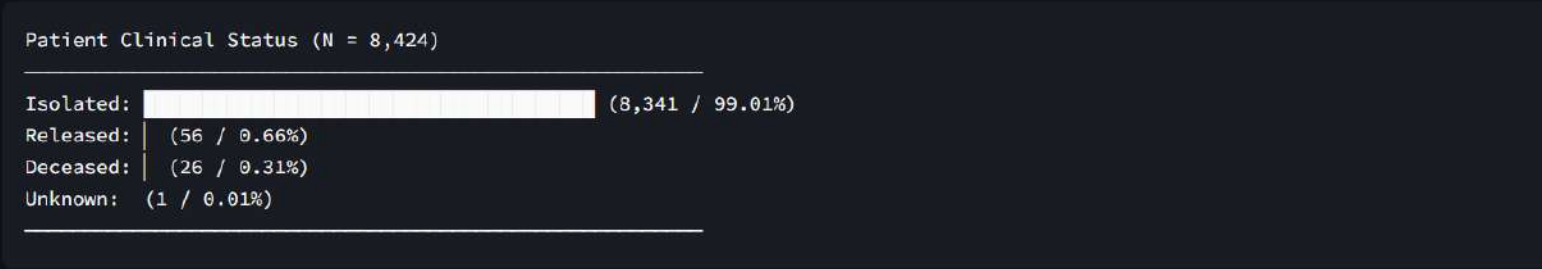
- **Epicenter Status:** With 99.77% of the cohort located in South Korea, this is a domestic crisis.
- **Intra-National Blindspot:** The `region` field is missing for 92.75% of cases (7,813 missing). From a resource-planning perspective, this is highly dangerous. Public health command centers cannot determine whether these 8,405 cases are concentrated in a single dense urban center (e.g., Seoul) or a specific province (e.g., Daegu), severely hindering the targeted deployment of ventilators, ICU beds, and personal protective equipment (PPE).

International Importation Vectors

- **Primary Lines of Introduction:** The presence of cases from China (16) and Mongolia (2) suggests early-stage importation. While these numbers are currently small, they represent critical nodes for border screening and international travel restrictions to prevent continuous seeding of new clusters.

4. Operational Insights & Actionable Recommendations

The extremely high proportion of active isolations (99.0%) combined with severe data missingness requires an immediate, dual-track response: **Clinical Surge Preparation** and **Information Architecture Restoration**.



Recommendation 1: Prepare for a Delayed Severe Outflow (Clinical Capacity Scaling)

- The Illusion of Low Mortality:** Currently, the crude case fatality rate (CFR) is 0.31% (26 deaths out of 8,424 cases). However, because 99% of the cohort is still isolated, their clinical outcomes have not yet been determined.
- Action:** Operationalize a 3-week clinical lag model. Assuming a standard 10–15% hospitalization rate for moderate-to-severe symptoms, prepare healthcare systems for an imminent wave of up to 1,200 concurrent hospitalizations and approximately 150–250 ICU admissions over the next 14 to 21 days.

Recommendation 2: Transition from Forward to Backward (Cluster) Contact Tracing

- **The Reality:** Traditional forward contact tracing (finding everyone a patient exposed) is impossible with a 99.2% missingness rate in `contact_number`.
- **Action:** Pivot epidemiology teams to "backward tracing." Focus exclusively on identifying the 1.82% of populated `group` and 3.10% of `infection_reason` data to locate common exposure sites (venues, events). Close down high-risk venues immediately rather than trying to track down individual contacts.

Recommendation 3: Implement Automated Digital Patient Intake (Resolving the "Data Fog")

- **The Reality:** Manual interviews by public health staff have failed to keep pace with the outbreak, leading to >92% missing data on vital risk factors like age (`birth_year`) and biological sex.
- **Action:** Deploy an automated, SMS-based or mobile web patient portal linked to the `id` generated upon a positive laboratory test. Patients must self-report their `birth_year` , `sex` , `region` , and basic symptom onset details before their isolation protocol is finalized. This will instantly populate the database, enabling demographic risk-stratification (e.g., identifying elderly patients at high risk of mortality).

Recommendation 4: Stratified Isolation Strategy

- **The Reality:** Isolating 8,341 individuals within traditional hospital infrastructure is unsustainable and will lead to immediate system collapse.
- **Action:** Implement a triaging framework:
 - **Home Isolation:** For asymptomatic/mild cases under 50 with no comorbidities.
 - **Community Treatment Centers:** Repurpose dormitories/hotels for mild cases requiring basic monitoring, offloading hospitals.
 - **Tertiary Hospitals:** Strictly reserve ICU beds and negative pressure rooms for patients over 65 or those with severe respiratory distress.

5. Conclusion

This dataset captures a classic "fog of war" scenario in early-stage epidemic management. While the raw clinical metrics show a low mortality rate, this is highly deceptive: it is a function of a young epidemic curve where 99% of cases are still active and isolated.

The critical threat to the response is not just the virus, but the **failure of the epidemiological data pipelines**. With contact tracing, demographic profiling, and regional tracking missing for over 92% of the cohort, the response team is currently operating without situational awareness.

By immediately deploying automated digital intake to resolve demographic data gaps, shifting to cluster-based backward contact tracing, and preparing clinical infrastructure for a lag-driven wave of severe cases, public health authorities can regain control of the transmission curve and mitigate preventable mortality.



Interactive AI Analyst Desk

Ask a professional question about patterns, demographics, or sources:

Ask AI Workspace

AI Workflow Completed Successfully



COVID-19 Analytics Dashboard

Exploratory visualizations and clinical pattern distribution desks.

✔ Dataset Loaded and Normalized Successfully



Dataset Preview

	id	sex	birth_year	country	region	group	infection_reason	infection_order	infected_by	contact_number	confirmed_date	released_date
0	1	female	1984	China	filtered at airport	None	visit to Wuhan	1	None	45	2020-01-20	2020-02-06
1	2	male	1964	Korea	filtered at airport	None	visit to Wuhan	1	None	75	2020-01-24	2020-02-05
2	3	male	1966	Korea	capital area	None	visit to Wuhan	1	None	16	2020-01-26	2020-02-12
3	4	male	1964	Korea	capital area	None	visit to Wuhan	1	None	95	2020-01-27	2020-02-09
4	5	male	1987	Korea	capital area	None	visit to Wuhan	1	None	31	2020-01-30	None
5	6	male	1964	Korea	capital area	None	contact with patient	2	3	17	2020-01-30	2020-02-19
6	7	male	1991	Korea	capital area	None	visit to Wuhan	1	None	9	2020-01-30	2020-02-15
7	8	female	1957	Korea	Jeollabuk-do	None	visit to Wuhan	1	None	113	2020-01-31	2020-02-12

Total Rows Logged

8424

Total Data Features

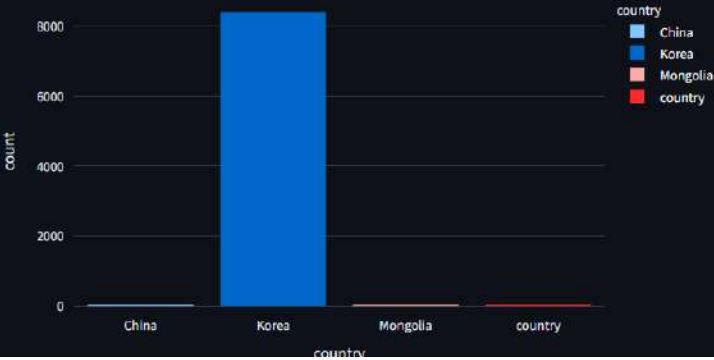
14

Total Missing Cells

81648

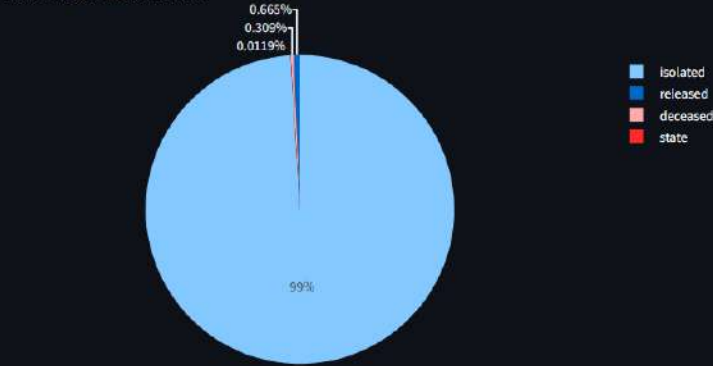
Country-wise Patient Distribution

Case Counts by Country



Patient Outcome Breakdown

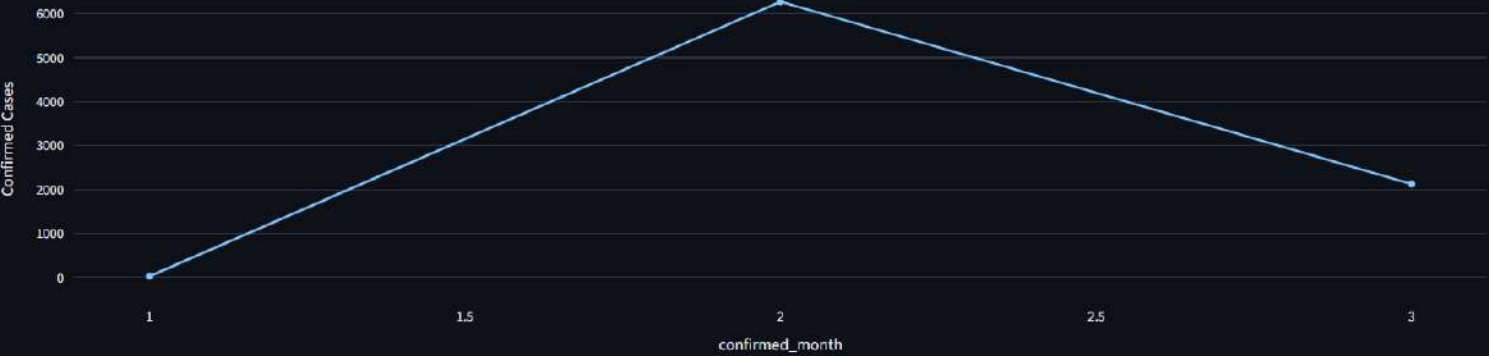
Patient Resolution States



 Line Analysis: Cases Logged Across Month Timelines

Timeline Case Trends By Month Number



Feature Matrix Correlations

Not enough numerical dimensions are available to construct a valid covariance matrix map.

Interactive Variable Distribution

Select Target Attribute to Profile:

confirmed_month

🔍 + 📊 📄 🗑️ ↺

Density Breakdown: confirmed_month



📄 Export Cleaned Dataset (CSV)



COVID-19 Patient State Prediction

Predict the patient's probable recovery trajectory using safe, optimized ML model pipelines.

Patient Information Profile

Patient Location Country

Korea



Confirmed Month

3



Confirmed Year

2020



Confirmed Day

15



Configured Attribute Input Summary

Clinical Attribute Feature	Input Parameter Value
Patient Country Location	Korea
Case Confirmed Year	2020
Case Confirmed Month	3
Case Confirmed Day	15

 [Evaluate Target Recovery Odds](#)